

Chapter 50 — Renal Dysgenesis and Cystic Disease of the Kidney — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Dysplasia** = histological; **hallmark primitive duct**; commonest cause of childhood ESRD (end-stage renal disease).
- **Autosomal dominant polycystic kidney disease (ADPKD)**: PKD1 (chromosome 16, 85%, ESRD 50s) vs PKD2 (chromosome 4, 15%, ESRD 70s).
- **Ravine**: <30 y ≥ 2 cysts; 30–59 y ≥ 2 each kidney; ≥ 60 y ≥ 4 each.
- **ADPKD: RCC (renal cell carcinoma) not increased**; berry aneurysm 10–30%.
- **Autosomal recessive PKD (ARPKD)**: chromosome 6 (PKHD1); congenital hepatic fibrosis always.
- **Tuberous sclerosis (TSC) angiomyolipoma (AML)** >3 cm, growing → mammalian target of rapamycin (mTOR) inhibitor (everolimus) first line.
- **von Hippel–Lindau (VHL)**: 3p25; RCC about 50%; nephron-sparing surgery = standard (control, not cure).
- **Multicystic dysplastic kidney (MCDK)**: commonest cystic disease; 40% involute; no hypertension or cancer risk.
- **Bosniak**: I–II no surgery; IIF follow (11% progress); III explore; IV nephrectomy.
- **Acquired renal cystic disease (ARCD)**: >90% after 10 years' dialysis; tumour >3 cm → excise.

OVERVIEW OF GENETICS

- **Genes**: PAX2 → hypodysplasia; EYA1 → branchio-oto-renal; HNF1B → cysts + diabetes.

RENAL AGENESIS AND DYSPLASIA

- **Dysplasia**: primitive ducts, cartilage; primary (genetic) or secondary (obstruction).
- **Ultrasound dysplasia**: small, brighter than liver, lost corticomedullary differentiation.

HYPOPLASIA AND HYPODYSPLASIA

- **Degree of dysplasia** $\propto$ degree of ureteric ectopia.
- **Lateral ectopia** → reflux; **medial ectopia** ($\pm$ ureterocele) → hypodysplasia.
- **Oligomeganephronia**: fewer, bigger nephrons; bilateral; proteinuria first; ESRD second decade.
- **Oligomeganephronia treatment**: protein 1.5 g/kg/day; ACE (angiotensin-converting enzyme) inhibitor even if normotensive.
- **Oligomeganephronia not familial** → living related donor acceptable.
- **Ask-Upmark**: young females, ≥ 10 y; **severe hypertension**; midzone slit-scar; 12–35 g.
- **Ask-Upmark: nephrectomy cures hypertension regardless of renin**; reflux repair does not.
- **Posterior urethral valves**: mild (subcapsular cysts) vs severe (large cysts, cartilage) hypodysplasia.

CYSTIC DISEASES OF THE KIDNEY

- **Multicystic** = dysplastic; **polycystic** = normal development, no dysplasia.
- **Cyst biology**: proliferation ("second hit"), fluid secretion, matrix change; **primary cilium**.
- **Chromosome 16** = PKD1 + TSC2 → contiguous gene syndrome.

AUTOSOMAL RECESSIVE (INFANTILE) POLYCYSTIC KIDNEY DISEASE

- **1:10,000–50,000** (table 1:5000–40,000); up to 50% die in first days.
- About half of neonatal survivors alive at 10 years.
- **Ultrasound**: huge, **homogeneous hyperechoic** kidneys; microcysts; pyramids blend.
- **Older survivors**: portal hypertension, varices; liver enzymes normal.

- **Very early bilateral nephrectomy** (<3 months) → worst neurological outcome (ARegPKD registry).
- **No renal neoplasm association**.

AUTOSOMAL DOMINANT (ADULT) POLYCYSTIC KIDNEY DISEASE

- **1:400–1000**; commonest inheritable; 7–15% of haemodialysis; 10% sporadic.
- **96% manifest by 90**; renal failure rare <40.
- **Symptoms 30–50 y**; pain = commonest adult presentation.
- **Haematuria 50%**; stones 20–30% (uric acid = calcium oxalate).
- **Anaemia unusual** (erythropoietin raised).
- **Hypertension**: 50% at 20–35 y; >60% before impairment; ~100% at ESRD; renin-mediated.
- **Early ESRD risk**: PKD1, male, haematuria <30 y, hypertension <35 y, sickle trait.
- **Liver cysts: commonest extrarenal**; almost all by 50; worse in women.
- **Seminal vesicle cysts 40%**; arachnoid 8%; pancreas 5%.
- **Subarachnoid haemorrhage (SAH) death** about 9% of those with aneurysm.
- **RCC in ADPKD**: bilateral 12%, multicentric 28%, sarcomatoid 33%.
- **PKD2 <30 y**: ultrasound sensitivity only 67% → genetic testing.
- **No family history**: bilateral cysts + ≥ 2 of (enlargement, ≥ 3 liver cysts, aneurysm, other cyst).
- **Acute cyst haemorrhage 50–90 HU** (Hounsfield units) on CT (computed tomography).
- **First-line antihypertensive**: ACE inhibitor or ARB (angiotensin receptor blocker); target <130/80.
- **Avoid NSAIDs** (non-steroidal anti-inflammatory drugs); limit opioids.
- **Cyst infection**: lipophilic antibiotics — trimethoprim-sulfamethoxazole, fluoroquinolone, chloramphenicol.
- **Unroofing relieves pain only**; nephrectomy for symptomatic ESRD.
- **Screen for aneurysm only if**: family history, prior rupture, major surgery, pilot, anxiety.
- **Aneurysm <7 mm** → observe; >7–10 mm or symptomatic → clip or coil.
- **HALT-PKD**: ACE inhibitor + ARB = ACE inhibitor alone; rigorous BP (blood pressure) slows kidney growth.
- **TEMPO/REPRISE**: tolvaptan (V2 antagonist) slows growth and decline; aquaresis, liver toxicity.

JUVENILE NEPHRONOPHTHISIS AND MEDULLARY CYSTIC DISEASE COMPLEX

- **Juvenile nephronophthisis (NPH):** recessive; **10–20% of childhood renal failure**; ESRD mean 13, <25.
- **Medullary cystic kidney disease (MCKD):** dominant; ESRD third–fourth decade; **hypertension**; type 2 gout.
- **Both:** polyuria/polydipsia >80%, vasopressin-resistant salt wasting; corticomedullary cysts.
- **Senior-Løken:** NPH + retinitis pigmentosa (16%).
- **Treat NPH:** replace sodium; avoid salt restriction and diuretics.

MULTIPLE MALFORMATION SYNDROMES WITH RENAL CYSTS

- **TSC:** up to 1:6000; TSC1 chromosome 9, TSC2 chromosome 16.
- **TSC triad:** epilepsy 80%, intellectual disability 60%, adenoma sebaceum 75%.
- **Earliest TSC skin sign:** ash-leaf macule; ultraviolet light.
- **TSC diagnosis:** 2 major, or 1 major + 2 minor; renal AML major, renal cysts minor.
- **TSC AML 40–80%;** rare <6 y, common >10 y; never metastasises.
- **TSC RCC about 2%** (table 3%); female, young, bilateral.
- **Suspect RCC in TSC:** enlarging fat-free or calcified lesion.
- **TSC imaging:** magnetic resonance imaging (MRI) preferred (25–30% AML fat-poor); contrast safe if GFR (glomerular filtration rate) ≥60.
- **AML second line:** embolisation + steroids; resection; ablation. **Acute bleed:** embolise + steroids.
- **TSC: avoid nephrectomy.** Image every 1–3 years; yearly if near 3 cm.
- **VHL:** 1:30,000–50,000; >95% penetrance by 65; mean age 35–40.
- **VHL:** renal cysts 76% (bilateral 75%); pheochromocytoma 10–17%; pancreatic cysts up to 70%.
- **VHL epididymal cystadenoma 10–26%** — benign; observe.
- **Cerebellar haemangioblastoma:** most characteristic; over half of VHL deaths.
- **VHL diagnosis:** family history + 1 lesion; none → ≥2 haemangioblastomas or 1 + another.
- **VHL surgery:** recurrence 75–85%, 10-year disease-specific survival 81–94%.
- **VHL pheochromocytoma → partial adrenalectomy** (recurrence 15%).
- **VHL Alliance 2020:** eye <1 y; brain, spine MRI + audiogram 2-yearly from 11; abdominal MRI 2-yearly from 15.
- **VHL: screen only genetic carriers.**

MULTICYSTIC DYSPLASTIC KIDNEY

- **MCDK 1:1000–4000; male 55–70%;** contralateral VUR (vesicoureteric reflux) 18–43%.
- **Contralateral pelviureteric junction (PUJ) obstruction 3–12%.**
- **MCDK vs PUJ obstruction:** MCDK has no central pelvis, no communication, no function.
- **Remove MCDK only if:** mass effect, hypertension, rupture, increasing solid tissue.

BENIGN MULTILOCLAR CYST (CYSTIC NEPHROMA)

- **Multilocular cyst:** 95% <4 y (boys 2:1) or >30 y (women 8:1).
- **Multilocular cyst → nephrectomy (partial if possible);** diagnosis only by excision.

SIMPLE CYSTS

- **Simple cysts:** children 0.22%; 20% by 40; 50% after 60.
- **Ultrasound:** no echoes, thin smooth wall, acoustic enhancement, spherical.
- **CT:** –10 to +20 HU, no enhancement → about 100% accuracy.
- **Hyperdense cyst benign if:** 20–90 HU, no enhancement, ≤3 cm, ≥¼ exophytic.
- **Calcification:** cyst 2% (peripheral) vs RCC 10% (central).
- **Cyst puncture only if:** infection; ultrasound echoes but classic CT; poor surgical candidate.
- **Child:** simple cyst <15 mm → repeat ultrasound at 2 years, then discharge.
- **Paediatric Bosniak III–IV:** >90% intermediate or malignant.

MEDULLARY SPONGE KIDNEY

- **Medullary sponge kidney (MSK):** "brush bristles"; colic 50–60%, UTI (urinary tract infection) 20–33%.
- **MSK: hypercalciuria one-third to one-half; incomplete distal renal tubular acidosis 30–40%.**
- **MSK first line:** potassium citrate; thiazide for stone formers even if normocalciuric.

SPORADIC GLOMERULOCYSTIC KIDNEY DISEASE

- **Sporadic glomerulocystic disease:** no family history, no anomalies.

ACQUIRED RENAL CYSTIC DISEASE

- **ARCD: ≥3 cysts per kidney;** 10% pre-dialysis → 44% at 3 y → 60% at 5 y.
- **ARCD: uraemia, not dialysis, is the cause; male 3:1.**
- **ARCD kidneys <300 g vs ADPKD >800 g.**
- **Dialysis renal malignancy 5–50%;** post-transplant RCC >15%.
- **>70% of transplant RCCs arise in native kidneys.**
- **ARCD screening:** after 3 years' dialysis; <55 y, good health.
- **Retroperitoneal bleed, RCC not excluded → nephrectomy.**
- **Gadolinium in ESRD → nephrogenic systemic fibrosis risk.**

PARAPELVIC AND RENAL SINUS CYSTS

- **Renal sinus cysts:** 15–20% at autopsy; mimic hydronephrosis; conservative.

CLASSIC TRAPS

- **Trap:** multicystic = polycystic — **Actually:** multicystic is dysplastic; polycystic is not.
- **Trap:** ADPKD raises RCC incidence — **Actually:** not increased; ARCD does.
- **Trap:** juvenile NPH causes hypertension — **Actually:** only MCKD does.
- **Trap:** salt restriction in NPH — **Actually:** replace sodium; avoid diuretics.
- **Trap:** neonatal echogenic kidneys = ARPKD — **Actually:** ADPKD can mimic; scan parents.
- **Trap:** negative Ravine scan <30 excludes PKD2 — **Actually:** sensitivity only 67%.
- **Trap:** TSC AML >4 cm → embolise — **Actually:** >3 cm growing → mTOR inhibitor.
- **Trap:** VHL surgery aims at cure — **Actually:** control, spare nephrons.
- **Trap:** calcification makes a cyst malignant — **Actually:** enhancement does (IIF calcified).
- **Trap:** Ask-Upmark hypertension cured by reflux repair — **Actually:** needs nephrectomy.